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DAA LAB

LAB EXPERIMENTS

EXPERIMENT – 2

CHAMPAGNE TOWER

AIM:

To implement a Python program to determine the amount of champagne present in a particular glass of a champagne tower after pouring a specified number of cups.

ALGORITHM:

1. Initialize a two-dimensional array representing the champagne glasses.
2. Pour the given amount of champagne into the top glass.
3. For each row, check the amount of champagne in every glass.
4. If a glass contains more than one cup, calculate the excess amount.
5. Divide the excess equally between the two glasses directly below it.
6. Continue the process row by row until the required query glass is reached.
7. The amount in the requested glass is limited to a maximum of one cup.
8. Return and display the amount of champagne in the required glass.

PROCEDURE:

1. Open a Python environment such as Google Colab, Jupyter Notebook, or IDLE.
2. Define a function named champagne_tower().
3. Enter the number of poured cups, query row, and query glass.
4. Store the champagne distribution in a two-dimensional list.
5. Run the program to distribute the excess champagne to the next row.
6. Display the amount present in the requested glass.

PYTHON CODE:

```python
def champagne(poured, row, glass):
    tower = [[0] * 10 for _ in range(10)]
    tower[0][0] = poured

    for i in range(row):
        for j in range(i + 1):
            if tower[i][j] > 1:
                extra = (tower[i][j] - 1) / 2
                tower[i][j] = 1
                tower[i + 1][j] += extra
                tower[i + 1][j + 1] += extra

    return min(1, tower[row][glass])


print(champagne(4, 2, 1))
```

SAMPLE OUTPUT:

Number of paths : 4

```
Output

0.5
```

RESULT:

Thus, the Python program to find the number of paths that move out of the boundary was successfully implemented and executed.